Carlos Eduardo Cancino-Chacón

PERSONAL DETAILS

Birth	June 6, 1986
Citizenship	Austrian, Mexican
Address	Institute of Computational Perception Johannes Kepler University Linz Wiesingerstrasse 4, Second Floor, Room 250 A-1010, Vienna, Austria
E-Mail	carlos.cancino_chacon@jku.at
Website	http://www.carloscancinochacon.com
GitHub	https://github.com/CarlosCancino-Chacon
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MAIN AREAS OF RESEARCH

- Computational Music Rehearsal Analysis
- Computational models of expressive music performance
- Real-time human-computer interaction in music performance (automatic accompaniment systems, interfaces for musical expression)
- Cognitively-plausible computational models of music analysis and music cognition
- Music Information Retrieval
- Machine learning (deep learning, probabilistic models)

ACADEMIC QUALIFICATIONS

Doctoral degree in Computer Science 10/2014–12/2018
Johannes Kepler University of Linz, Austria
Supervisor: Gerhard Widmer
Co-supervisor: Maarten Grachten
Thesis: Computational Modeling of Expressive Music Performance Through Linear and Non-linear Basis Function Models

Master's degree in Electrical Engineering and Audio Engineering 10/2011–07/2014
Graz University of Technology/University of Music and Performing Arts Graz, Austria
Supervisor: Franz Pernkopf
Thesis: Tarkus Belief Propagation: On Message Passing Algorithms and Computational Commutative Algebra

Undergraduate degree in Physics 08/2005–03/2011
National Autonomous University of Mexico, Mexico City, Mexico
Supervisor: Marcos Ley Koo
Thesis: Análisis teórico experimental de transductores de ultrasonido tipo Langevin (*Theoretical–experimental analysis of ultrasonic Langevin transducers*)

Undergraduate degree in Piano Performance 09/1999–02/2011
National Conservatory of Music, Mexico City, Mexico
Supervisor: Héctor Alfonso Rojas Ramírez

RESEARCH EXPERIENCE

Senior Scientist 10/2025-present
Institute of Computational Perception
Johannes Kepler University Linz, Austria

Assistant Professor (Universitätsassistent) 11/2020-9/2025
Institute of Computational Perception
Johannes Kepler University Linz, Austria

Guest Researcher 01/2020-10/2021
RITMO Centre for Interdisciplinary Studies in Rhythm, Time and Motion
University of Oslo, Norway
Project: MIRAGE

Postdoctoral Researcher 12/2018-10/2020
Austrian Research Institute for Artificial Intelligence, Vienna, Austria
Intelligent Music Processing and Machine Learning Group
Project: Con Espressione
Supervisor: Gerhard Widmer

Doctoral Researcher 01/2014-12/2018
Austrian Research Institute for Artificial Intelligence, Vienna, Austria
Intelligent Music Processing and Machine Learning Group
Projects:

- Lrn2Cre8 (01/2014 – 09/2016)
  Supervisors: Maarten Grachten and Gerhard Widmer
- Con Espressione (10/2016 – 12/2018)
  Supervisor: Gerhard Widmer

GRANTS

Dress Rehearsal: Towards Computer-Aided Music Rehearsal
Austrian Science Fund (FWF) ASTRA Award 2027-2032
Principal Investigator (grant number AST 2939225)
Total funding: €1,000,000.00

AURA: Augmenting musical interaction via Embodied Virtual Avatars
Weave: Research Foundation–Flanders (FWO) and Austrian Science Fund (FWF) 2025-2029
Principal Investigator (FWO-leading agency) ***Pieter-Jan Maes*** *(grant number G0A3R25N)*
Principal Investigator (FWF) ***Carlos Cancino-Chacón*** *(grant number PIN 1347924)*
FWF share: €264,679.00 *(total Weave project budget:* €826,979.00*)*

Rach3: A Computational Approach to Study Piano Rehearsals
Austrian Science Fund (FWF) 2024-2027
Principal Investigator (grant number PAT 8820923)
Total funding: €457,812.00

TEACHING AND SUPERVISION

External Lecturer 03/2023-10/2023
University of Music and Performing Arts Vienna, Austria

- Musical Computing (PhD level)

Assistant Professor 10/2020-present
Johannes Kepler University Linz, Austria

- Musical Informatics (undergraduate/master's level)
- Reinforcement Learning (undergraduate level)
- Machine Learning and Pattern Classification (exercise track; undergraduate)
- Seminars in Data Science and Artificial Intelligence (undergraduate and master's level)
- Artificial Intelligence (exercise track; undergraduate level)

Course Lecturer 01/2011-07/2011
National Conservatory of Music, Mexico City, Mexico

- Elementary Music Theory I
- Harmony (Levels I-III)

Supervision/Co-Supervision
Johannes Kepler University Linz, Austria

Current

- **Alexander Neuhauser** (November 2025-present) PhD Co-Supervisor (Project PI). Main supervisor: Gerhard Widmer.
- **Ivan Pilkov** (October 2024-present) PhD Co-Supervisor (Project PI). Main supervisor: Gerhard Widmer.
- **Suhit Chirithapudi** (November 2024-present) PhD Co-Supervisor (Project PI). Main supervisor: Gerhard Widmer.
- **Fabio Demko** (Bachelor's thesis, expected Early 2027).
- **Arian Moradi** (Bachelor's thesis, expected Late 2026).

Finished

- **Huan Zhang** (June 2022-December 2025) External PhD Mentor as part of the AIM Mentoring Programme at Queen Mary University of London (main supervisor: Simon Dixon).
- **Julian Bakan** (Bachelor's thesis, November 2025) A Systematic Comparison of Dynamic Time Warping and Hidden Markov Models for Symbolic Score-to-Performance alignment.
- **Alexander Neuhauser** (Master's thesis, October 2025) Difficulty Estimation and Robotic Piano Performance via Reinforcement Learning
- **Kseniia Leonteva** (Bachelor's thesis, October 2025) Predicting the Electrical Demand of Aluminium Hot Rolling
- **Daryna Vorona** (Bachelor's thesis, September 2025) Convolutional Neural Networks for Difficulty Classification of Piano Sheet Music
- **Adam Štefunko** (February 2025-July 2025 Internship as part of PhD program). Symbolic Alignment for Basso Continuo Performances. Main PhD supervisor: Jan Hajič jr.

- **Maximilian Hofmann** (Bachelor's thesis, February 2025). Transformer models for Expressive Piano Performance Rendering
- **Ivan Pilkov** (Master's Thesis, September 2024). Automatic Pose Estimation for Kinematic Analysis of the Arm-Hand-Fingers Chain of Pianists.
- **Alexander Deutschbauer** (Bachelor's thesis, August 2024). TPDD: A Dataset for Automatic Difficulty Estimation of Solo Piano Pieces.
- **Uros Zivanovic** (Bachelor's thesis, September 2023). Transformers for Piano Playing Detection and Note Detection from Video.
- **Jasmin Roll** (Bachelor's thesis, September 2023). Evaluating Real-Time Audio-to-Audio Alignment using OLTW and HMM on Piano and Violin Music.
- **Simon Bauer** (Master's Project, July 2023). Computer-Generated Music for Traditionally Non-Musical Story Experiences.
- **CoPeWithUs Team** (March 2022- June 2022) Co-supervised (with Silvan Peter) a team of 4 master's students (Anna Hausberger, Christoph Pichler, Ivan Pilkov and Jakob Wögerbauer) who participated in the AI Song Contest 2022.

Austrian Research Institute for Artificial Intelligence, Vienna, Austria

- **Simon Bahadoran** (Summer Internship/Master's Thesis Summer 2020). Development of an encoder-decoder model for conditional prediction of expressive piano performance.
- **Martin Bonev** (2017-2019). Development of a polyphonic score follower for an expressive accompaniment system.
- **Federico Simonetta** (Summer Internship Summer 2018). Main melody line identification from symbolic scores using convolutional neural networks.

ACADEMIC ACTIVITIES

ACADEMIC COMMITTEES

- Secretary of the Board of the International Society for Music Information Retrieval for the period 2026-2027.
- Member-at-large of the Board of the International Society for Music Information Retrieval for the period 2024-2025
- Member of the Editorial Board of the Transactions of the International Society for Music Information Retrieval.
- Co-organizer, LatAm BISH Bash Meetup group.
- Scientific Program Chair for the 2nd Latin American Music Information Retrieval Workshop.
- Scientific Program Chair for the 1st Latin American Music Information Retrieval

REVIEWING

- **International Scientific Journals**
  - Journal of New Music Research, Neural Computing and Applications, Applied Sciences, Transactions of the International Society for Music Information Retrieval, Frontiers in Psychology, Music Perception, Music & Science.

- **International Conferences**
  - International Society for Music Information Retrieval Conference, International Joint Conference on Artificial Intelligence, Sound & Music Computing Conference, Biennial International Conference on Mathematics and Computation in Music, International Workshop on Deep Learning for Music, Multimedia Modeling Conference

INVITED SCIENTIFIC TALKS AND TUTORIALS

1. Cancino-Chacón, C. (September 2026) "Towards Computational Piano Rehearsal Analysis: The Rach3 Project", Keynote Talk presented at the 7th International Workshop on Reading Music Systems (WoRMS 2026), Vienna, Austria.

2. Cancino-Chacón, C. (July 2026) "Ensayo General: Usando Inteligencia Artificial para Entender la Interpretación Musical", invited talk presented at the Universidad Anáhuac, Mexico City, Mexico.

3. Bömcke-Vollmer, F., Peter, S., Hu, P., Jiang, Y., Cancino-Chacón, C., Goebl, W., Engeln, L., Münzmay, A., Hentschel, J., Weigl, D. M. and Berndt, A. (May 2026) "Approaches to Musical Performance Research: Techniques, Formats, Software", Workshop presented at the Music Encoding Conference 2026, Tokyo, Japan.

4. Cancino-Chacón, C., Park, J., Peter, S., Özer, Y. and Chirithapudi, S. (May 2026) "Music Alignment Uncovered: Representations, Algorithms and Hands-On Tools", Workshop presented at the Music Encoding Conference 2026, Tokyo, Japan.

5. Cancino-Chacón, C. (May 2026) "An AI Dress Rehearsal: Exploring Music Performance and Interaction with Computational Models", invited talk presented at Yamaha's MINALab, Yokohama, Japan.

6. Cancino-Chacón, C. (May 2026) "An AI Dress Rehearsal: Exploring Music Performance and Interaction with Computational Models", invited talk at the Lecture Series Music, Art, and Mathematics / University Salzburg and Mozarteum University Salzburg, Austria.

7. Cancino-Chacón, C. (January 2026) "Künstliche Intelligenz in der Musik", invited talk presented at the Taktwechsel 2026 Event organized at Klavierhaus Förstl in Vienna, Austria.

8. Cancino-Chacón, C. (December 2024) "An AI Dress Rehearsal: Exploring Music Performance and Interaction with Computational Models", Keynote Talk presented at the 1st Latin American Music Information Retrieval Workshop, Rio de Janeiro, Brazil.

9. Cancino-Chacón, C. (December 2023) "Towards Expressive Artificial Musical (Co-)performers", invited talk at the 5th LatAm BISH Bash, online.

10. Cancino-Chacón, C. (December 2023) "Towards Expressive Artificial Musical Co-performers: The ACCompanion Story", invited talk at the International Symposium on AI and Music Performance, Daejeon, South Korea.

11. Cancino-Chacón, C. (June 2023) "Towards Understanding Emotion Communicated Through Performance of Orchestral Music: Preliminary Results", invited talk at the 2nd MIRAGE Symposium, RITMO Centre for Interdisciplinary Studies in Rhythm, Time and Motion, University of Oslo, Norway.

12. Cancino-Chacón, C., Peter, S., Karystinaios, E. and Foscarin, F. (December 2022) "An Introduction to Symbolic Music Processing with Partitura", tutorial session at the 23rd International Society for Music Information Retrieval Conference, Bengaluru, India.

13. Cancino-Chacón, C. (September 2022) "Play it Again, Hall 9000! Towards Expressive Computational Music Performers", invited talk at the Max Planck Institute for Intelligent Systems, Tübingen, Germany.

14. Cancino-Chacón, C. (March 2022) "Decoding Communicated Emotion in Expressive Music Performance (of the Rach 3!)", invited talk at the MIRAGE Workshop, University of Oslo, Norway.

15. Lartillot, O., Guldbransen, E. and Cancino-Chacón, C. E. (June 2021) "Dynamics analysis, and application to a comparative study of Bruckner performances", invited talk at the Mirage Symposium #1: Computational Musicology, RITMO Centre for Interdisciplinary Studies in Rhythm, Time and Motion, University of Oslo, Norway.

16. Cancino-Chacón, C. (April 2020) "I'll be Bach! Modeling Expressive Performance with Machine Learning", invited talk at the Food & Paper Talk Series, RITMO Centre for Interdisciplinary Studies in Rhythm, Time and Motion, University of Oslo, Norway.

17. Cancino-Chacón, C. (March 2020) "Machine Listening of Orchestral Music", invited talk at the Workshop on Musical Listening, University of Oslo, Norway.

18. Cancino-Chacón, C., Kosta, K. and Grachten, M. (November 2019) "Computational Models of Expressive Performance", tutorial session at the 20th International Society for Music Information Retrieval Conference, Delft, The Netherlands.

19. Cancino-Chacón, C. (March 2019) "Modeling Expressive Music Performance with Non-linear Basis Function Models", invited talk at the Deep Learning Seminar, University of Vienna, Austria.

20. Cancino-Chacón, C. (January 2019) "Computational Modeling of Expressive Music Performance with Linear and Non-linear Basis Function Models", invited talk at the Austrian Research Institute for Artificial Intelligence, Vienna, Austria.

21. Cancino-Chacón, C. (November 2016) "¿Escuchan los androides música electrónica?", invited talk at the Pláticas DeMentes talk series. Faculty of Psychology, National Autonomous University of Mexico.

22. Cancino-Chacón, C. (November 2016) "En busca del factor Mozart", invited talk at the National Conservatory of Music, Mexico City, Mexico.

SCOLARSHIPS AND AWARDS

Award for Creative Achievement 06/2017
AccompaniX Competition, 2017 Turing Tests in the Creative Arts.
$500 team award for development of an expressive computer accompaniment system.

Fundación INBA – CONACYT Scholarship
Mexican National Council for Science and Technology 10/2012-02/2014

PUBLIC OUTREACH AND DISSEMINATION

PUBLIC DEMONSTRATIONS

- *The Accompanion*, an AI-based Piano Accompaniment System.
  - (January 2026) Short Talk and Demo during the Taktwechsel 2026 Event organized at Klavierhaus Förstl in Vienna, Austria
    `https://www.klavierhaus-foerstl.at/taktwechsel`
  - (2024) Demo during Gerhard Widmer's Wiener Vorlesung: "Con Espressione! Über KI und das Menschliche in der Musik" in Vienna, Austria
    `https://www.youtube.com/live/uRfeNpoqVbo?si=PpoayCB-2xmZqlO3&t=3200`

  - (December 2023) Lecture Concert during the International Symposium on AI and Expressive Performance, KAIST, Daejeon, South Korea
    https://sites.google.com/view/isaimp2023/lecture-concert
  - (October 2022) Demo at the MAINS Matinee at the Heidelberg Forum Foundation, Heidelberg, Germany
    https://www.imaginary.org/node/2417
  - (July 2022) Demo during Gerhard Widmer's Keynote at IJCAI-ECAI 2022, Vienna, Austria
    https://youtu.be/HzrwBFPN618?t=3105
  - (July 2022) Demo at the Music Togetherness Symposium
    https://mdw.ac.at/togetherness/mt-symposium/mts-22-programme/
  - (November 2021) Demo at the Falling Walls Science Summit 2021, Berlin, Germany
    https://www.youtube.com/watch?v=KE6WhYxuWLk&t=799s
  - Playlist of videos of the ACCompanion recorded in Vienna.
    https://www.youtube.com/playlist?list=PLPUWmNCGflVNxPppflMNFMzWedlOhZePO

- *Con Espressione! Exhibit*, an interactive system designed for popular science exhibitions in collaboration with IMAGINARY gGmbH (https://www.imaginary.org/program/con-espressione). The exhibit has been presented at the following exhibits at the Heidelberg Forum Foundation in Heidelberg, Germany:
  - "La La Lab - The Mathematics of Music" (2019-2020)
    https://www.imaginary.org/exhibition/la-la-lab-the-mathematics-of-music
  - "I am A. I. in Heidelberg" (2022)
    https://www.imaginary.org/event/i-am-ai-in-heidelberg

INTERVIEWS AND PUBLIC PRESS

- Kang, M.-k. (2023). My piano teacher is an AI pianist who has performed with human performers. EDAILY [Published: Dec. 7th, 2023]. (In Korean) https://n.news.naver.com/article/018/0005633332?type=journalists
- Vižintin, A. (2023). Artificial Intelligence and Music. Radio Študent FM 89.3 [Broadcasted Oct. 8th, 2023]. https://radiostudent.si/znanost/zr-intervju/umetna-inteligenca-in-glasba
- Vincent, J. (2016). A night at the AI jazz club. The Verge [Published: Oct. 12th 2016]. https://www.theverge.com/2016/10/12/13247686/ai-music-composition-jazz-club-london-deep-learning

PUBLICATIONS

PEER REVIEWED PUBLICATIONS

1. Hentschel, J., Berndt, A., Cancino-Chacón, C., Dixon, S., Foo, A., Gotham, M., Hu, P., Köster, M., Martins, F. D., Mauro, D. A., Müller, M., Neuwirth, M., Pacha, A., Page, K. R., Peter, S., Polyakov, E., Pugin, L., Weigl, D. M., Weiß, C., and Widmer, G. (2026). Time to align! modelling musical timelines for music information retrieval and digital musicology. *Transactions of the International Society for Music Information Retrieval*, 9(1):384–404

2. Hu, P., Zhang, H., Peter, S. D., Cancino-Chacón, C. E., Dixon, S., and Widmer, G. (2026). Revisiting expressive timing in piano performance at scale: A distant listening, multi-corpus study. *Transactions of the International Society for Music Information Retrieval*, 9(1):329–346

3. Chiruthapudi, S., Štefunko, A., Peter, S. D., Hu, P., Hajič Jr., J., and Cancino-Chacón, C. (2026). A flexible encoding model for non-unique note alignments. In *Proceedings of the Music Encoding Conference 2026*, Tokyo, Japan

4. Štefunko, A., Cancino-Chacón, C., and Hajič Jr., J. (2026). On every note a Griff: Looking for a useful representation of Basso Continuo performance style. In *Proceedings of the Music Encoding Conference 2026*, Tokyo, Japan

5. Pilkov, I., Bishop, L., and Cancino-Chacón, C. (2026). Estimating pianists' hand and finger kinematics with markerless motion capture. In *Proceedings of the 10th International Conference on Movement and Computing*, MOCO '26, Montpellier, France. ACM

6. Neuhauser, A. and Cancino-Chacón, C. (2026). Estimating piano piece difficulty via embodied robotic hand performance analysis. In *Proceedings of the 10th International Conference on Movement and Computing*, MOCO '26, Montpellier, France. ACM

7. Chiruthapudi, S., Štefunko, A., Hajič jr., J., and Cancino-Chacón, C. E. (in press). Challenges in Basso Continuo Performance-to-Score Alignment. In *Proceedings of the 17th International Symposium on Computer Music Multidisciplinary Research (CMMR 2025)*, London, UK

8. Morsi, A., Chiruthapudi, S., Peter, S., Pilkov, I., Bishop, L., Maezawa, A., Serra, X., and Cancino-Chacón, C. (2025). Enabling Empirical Analysis of Piano Performance Rehearsal with the Rach3 MIDI Dataset. In *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)*, Daejeon, South Korea

9. Park, J., Cancino-Chacón, C., Chiruthapudi, S., and Nam, J. (2025). Matchmaker: An Open-Source Library for Real-Time Piano Score Following and Systematic Evaluation. In *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)*

10. Štefunko, A., Chiruthapudi, S., Hajič jr., J., and Cancino-Chacón, C. E. (2025). Basso Continuo Goes Digital: Collecting and Aligning a Symbolic Dataset of Continuo Performance. In *Proceedings of the 6th Conference on AI Music Creativity (AIMC 2025)*, Brussels, Belgium

11. Lartillot, O., Swarbrick, D., Upham, F., and Cancino-Chacón, C. E. (2025). Video Visualization of a String Quartet Performance of a Bach Fugue: Design and Subjective Evaluation. *Music & Science*, 8:20592043251352299

12. Høffding, S., Bergstrøm, R. J. F., Bishop, L., Lucas, P., Burnim, K., Cancino-Chacón, C., Clim, A., Good, M., Hansen, N. C., Karlsen, E. S., Kvale, L. H., Laeng, B., Lartillot, O., Lippert, E., Martin, R., Nielsen, N., Omprakash, R., Paulsrud, T. S., Rosas, F., Swarbrick, D., Sørbø, S., Upham, F., Vrasdonk, A., Vuoskoski, J., Wallot, S., Yi, W., Danielsen, A., and Jensenius, A. R. (2025). Introducing the MusicLab Copenhagen Dataset. *Music & Science*, 8:20592043241303288

13. Zivanovic, U., Pilkov, I., and Cancino-Chacón, C. E. (2025). Pay Attention to the Keys: Visual Piano Transcription Using Transformers. In *Proceedings of the 34th International Joint Conference on Artificial Intelligence*, Montreal, Canada

14. Hu, P., Marták, L. S., Cancino-Chacón, C., and Widmer, G. (2024). Towards musically informed evaluation of piano transcription models. In *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR 2024)*, San Francisco, CA, USA

15. Zhang, H., Chowdhury, S., Cancino-Chacón, C. E., Liang, J., Dixon, S., and Widmer, G. (2024). Dexter: Learning and controlling performance expression with diffusion models. *Applied Sciences*, 14(15)

16. Cancino-Chacón, C. and Pilkov, I. (2024). The Rach3 Dataset: Towards Data-Driven Analysis of Piano Performance Rehearsal. In *Proceedings of the 30th International Conference on Multimedia Modeling MMM24*, Amsterdam, Netherlands

17. Peter, S. D., Chowdhury, S., Cancino-Chacón, C. E., and Widmer, G. (2023c). Are we describing the same sound? An analysis of word embedding spaces of expressive piano performance. In *FIRE '23: Proceedings of the 15th Annual Meeting of the Forum for Information Retrieval Evaluation*, Panjim, India

18. Peter, S. D., Cancino-Chacón, C. E., Karystinaios, E., and Widmer, G. (2023b). Sounding out reconstruction error-based evaluation of generative models of expressive performance. In *Proceedings of the 10th International Conference on Digital Libraries for Musicology*, DLfM '23, pages 58–66, New York, NY, USA. Association for Computing Machinery

19. Zhang, H., Karystinaios, E., Dixon, S., Widmer, G., and Cancino-Chacón, C. (2023). Symbolic Music Representations for Classification Tasks: A Systematic Evaluation. In *Proceedings of the 24th International Society for Music Information Retrieval Conference (ISMIR 2023)*, Milan, Italy

20. Peter, S. D., Cancino-Chacón, C. E., Foscarin, F., McLeod, A. P., Henkel, F., Karystinaios, E., and Widmer, G. (2023a). Automatic Note-Level Score-to-Performance Alignments in the ASAP Dataset. *Transactions of the International Society for Music Information Retrieval*

21. Cancino-Chacón, C., Peter, S., Hu, P., Karystinaios, E., Henkel, F., Foscarin, F., and Widmer, G. (2023). The ACCompanion: Combining Reactivity, Robustness, and Musical Expressivity in an Automatic Piano Accompanist. In *Proceedings of the Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI-23)*, Macao, S. A. R

22. Cancino-Chacón, C. E. (2023). Commentary on A Computational Approach to the Detection and Prediction of (Ir)Regularity in Children's Folk Songs. *Empirical Musicology Review*, 6(2)

23. Cancino-Chacón, C., Peter, S. D., Karystinaios, E., Foscarin, F., Grachten, M., and Widmer, G. (2022b). Partitura: A Python Package for Symbolic Music Processing. In *In Proceedings of the Music Encoding Conference (MEC2022)*, Halifax, Canada

24. Foscarin, F., Karystinaios, E., Peter, S. D., Cancino-Chacón, C., Grachten, M., and Widmer, G. (2022). The match file format: Encoding Alignments between Scores and Performances. In *In Proceedings of the Music Encoding Conference (MEC2022)*, Halifax, Canada

25. Bishop, L., Cancino-Chacón, C., and Goebl, W. (2021). Beyond synchronization: How and why do ensemble performers communicate. In Timmers, R., Bailes, F., and Daffern, H., editors, *Together in Music: Participation, Co-Ordination, and Creativity in Ensembles*. Oxford University Press

26. Cancino-Chacón, C., Peter, S., Chowdhury, S., Aljanaki, A., and Widmer, G. (2020). On the Characterization of Expressive Performance in Classical Music: First Results of the *Con Espressione* Game. In *Proceedings of the 21th International Society for Music Information Retrieval Conference (ISMIR 2020)*, Montreal, Canada

27. Lartillot, O., Cancino-Chacón, C., and Brazier, C. (2020). Real-Time Visualization of Fugue Played by a String Quartet. In *Proceedings of the 17th Sound and Music Computing Conference (SMC2020)*, Torino, Italy

28. Bishop, L., Cancino-Chacón, C., and Goebl, W. (2019b). Moving to communicate, moving to interact: Patterns of body motion in musical duo performance. *Music Perception*, 37(1):1–25

29. Bishop, L., Cancino-Chacón, C., and Goebl, W. (2019a). Eye gaze as a means of giving and seeking information during musical interation. *Consciousness & Cognition*, 68:73–96

30. Simonetta, F., Cancino-Chacón, C., Ntalampiras, S., and Widmer, G. (2019). A Convolutional Approach to Melody Line Identification in Symbolic Scores. In *Proceedings of the 20th International Society for Music Information Retrieval Conference (ISMIR 2019)*, Delft, The Netherlands

31. Cancino-Chacón, C., Grachten, M., Goebl, W., and Widmer, G. (2018). Computational Models of Expressive Music Performance: A Comprehensive and Critical Review. *Frontiers in Digital Humanities*, 5:25

32. Velarde, G., Cancino-Chacón, C., Meredith, D., Weyde, T., and Grachten, M. (2018). Convolution-based classification of audio and symbolic representations of music. *Journal of New Music Research*, 47(3):191–205. doi: 10.1080/09298215.2018.1458885

33. Cancino-Chacón, C. E., Gadermaier, T., Widmer, G., and Grachten, M. (2017d). An Evaluation of Linear and Non-linear Models of Expressive Dynamics in Classical Piano and Symphonic Music. *Machine Learning*, 106(6):887–909

34. Grachten, M., Cancino-Chacón, C. E., Gadermaier, T., and Widmer, G. (2017). Towards computer-assisted understanding of dynamics in symphonic music. *IEEE Multimedia*, 24(1):36–46

35. Cancino-Chacón, C., Grachten, M., Sears, D. R. W., and Widmer, G. (2017c). What Were You Expecting? Using Expectancy Features to Predict Expressive Performances of Classical Piano Music. In *Proceedings of the 10th International Workshop on Machine Learning and Music (MML 2017)*, Barcelona, Spain

36. Cancino-Chacón, C., Grachten, M., and Agres, K. (2017b). From Bach to The Beatles: The Simulation of Human Tonal Expectation Using Ecologically-Trained Predictive Models. In *Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR 2017)*, Suzhou, China

37. Grachten, M. and Cancino-Chacón, C. E. (2017). Temporal dependencies in the expressive timing of classical piano performances. In Lessafre, M., Maes, P.-J., and Leman, M., editors, *The Routledge Companion to Embodied Music Interaction*, pages 360–369. Routledge

38. Velarde, Gissel and Weyde, Tillman and Cancino Chacón, Carlos and Meredith, David and Grachten, Maarten (2016). Composer Recognition Based On 2D-Filtered Piano Rolls. In *Proceedings of the 17th International Society for Music Information Retrieval Conference (ISMIR 2016)*, pages 116–121, New York, NY, USA

39. Gadermaier, T., Grachten, M., and Cancino-Chacón, C. E. (2016). Basis-Function Modeling of Loudness Variations in Ensemble Performance. In *Proceedings of the 2nd International Conference on New Music Concepts (ICNMC 2016)*, Treviso, Italy

40. Cancino Chacón, C. E. and Grachten, M. (2015). An Evaluation of Score Descriptors Combined with Non-linear Models of Expressive Dynamics in Music. In *Proceedings of the 18th International Conference on Discovery Science (DS 2015)*, pages 48–62, Banff, AB, Canada

41. Agres, K., Cancino, C., Grachten, M., and Lattner, S. (2015). Harmonics co-occurrences bootstrap pitch and tonality perception in music: Evidence from a statistical unsupervised learning model. In *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci 2015)*, Pasadena, CA, USA

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